A Spherical N-Series Area Law for Regular Refinement Families

A cubic obstruction under scale-dependent perturbation,
mixed-exponent recovery, and the classical even-power background

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Abstract

Regular refinement families for spherical area exhibit the extrapolation ladder of orders $2, 4, 6, 8$. This manuscript asks where that ladder ends. Our main result is a cubic obstruction: when mesh nodes are displaced by an amount proportional to the grid spacing, the even-power error expansion acquires an odd h^3 term with coefficient $\varepsilon C_2'(0)$, which the even-power operators cannot remove. They leave it with exact residual factors $-1/6$, $-1/90$ and $-1/810$ at two, three and four scales, so the extrapolated order stalls at three. For uniform square-cell grids split along a fixed diagonal and deformed by a fixed smooth, boundary-vanishing field, we prove the underlying even-power expansion uniformly in the displacement, give the cell coefficient in closed form, and certify by interval arithmetic that $C_2'(0) = -6.80779\ldots \times 10^{-4} \neq 0$ for the test field, so that for this family the third-order stall is a theorem. A deterministic numerical test on a chordally triangulated latitude–longitude cell confirms the prediction: the fitted cubic coefficient agrees with $\varepsilon C_2'(0)$, measured independently from fixed-displacement runs, to about three significant figures; the residual ratios approach the predicted $1/15$ and $1/9$; the extrapolated orders stall at three (three- and four-scale: 3.000); and a mixed-exponent operator built for the exponent set $\{2, 3\}$ restores fourth order. We give exact weights and first surviving moments for the mixed sets $\{2, 3\}$, $\{2, 3, 4\}$ and $\{2, 3, 4, 5\}$, and an elementary parity proof of the even-power expansion for spherical caps bounded by regular geodesic polygons. The ladder itself, for flat-triangle (chordal) triangulations of a fixed smooth parametrization, is classical: it was conjectured by Georg (1991) and proved by Verlinden and Cools (1994) and by Lyness (1994). We verify that the chordal spherical triangle, fan-refined polygon and uniform latitude–longitude cell of this study satisfy that theorem, give an independent direct proof for the latitude–longitude cell, and note that a 1994 experiment of Georg and Tausch coincides with our octant test. The scale weights are the classical Richardson–Romberg weights. The new proofs and the computer-assisted certificate were produced with AI assistance and await independent human review. Nothing is claimed for independent node jitter, adaptive meshes or arbitrary spherical meshes, and none of the classical ingredients is claimed as new.

1 Introduction

Many geometric area approximations admit structured error expansions in a refinement parameter $h \asymp N^{-1}$. When the errors are in even powers, the same approximation evaluated at successive scales can be combined to remove those terms, giving the sequence of orders

$$2 \longrightarrow 4 \longrightarrow 6 \longrightarrow 8.$$

For flat-triangle triangulations of a fixed smooth surface patch this structure is a theorem of the boundary-element literature (Section 5). The question studied here is what happens at the edge of

that theorem. In practice a mesh is often not the image of a uniform grid under one fixed map: nodes are smoothed, jittered or adapted by amounts tied to the local spacing. The perturbed-grid experiment of the original June 2026 manuscript showed the higher-order ladder deteriorating in exactly that situation while second-order convergence survived.

Section 6 gives a mechanism. A displacement $\delta = \varepsilon h$ turns the coefficient $C_2(\delta)$ of the even-power expansion into a function of h, and its first variation appears as an h^3 term. Because the even-power operators are built to annihilate h^2, h^4, h^6, they multiply an h^3 term by fixed nonzero factors, which we compute exactly. Theorem 2 proves the required expansion for smoothly deformed square-cell grids, and Proposition 2 certifies that the cubic coefficient is nonzero for the test field. A deterministic test, specified completely and included in the archive, confirms the amplitude law, the residual factors and the stall at third order. Section 7 gives the operators that remove a prescribed mixed set of exponents, and the same test confirms that the $\{2, 3\}$ operator restores fourth order.

The remaining sections record the fixed-map families: a spherical cap bounded by a regular geodesic polygon, for which we give a parity proof (Section 8); the uniform latitude–longitude cell, with an independent direct proof (Section 9); and the chordal spherical triangle and fan-refined polygon, which illustrate the classical theorem (Sections 10 and 11). Earlier versions of this manuscript were written without knowledge of the work of Georg [8], Georg and Tausch [9], Verlinden and Cools [17] and Lyness [12], and presented the triangle and polygon ladders as numerical evidence for a conjectural law. The original tabulated experiments are retained with their provenance limitations.

2 Contributions and scope

Results of this manuscript.

- *A cubic obstruction* (Proposition 1): under a joint even-power expansion in the mesh size and a displacement parameter, scale-dependent displacement produces an h^3 term $\varepsilon C_2'(0)h^3$, and the two-, three- and four-scale even-power operators leave residuals $-\frac{1}{6}$, $-\frac{1}{90}$, $-\frac{1}{810}$ times that term.

- *A displacement-uniform expansion* (Theorem 2): for uniform square-cell grids with a fixed diagonal, deformed by a fixed smooth boundary-vanishing field, the chordal area has an even-power expansion with smooth coefficients and a remainder uniform in the displacement. This supplies the hypothesis of Proposition 1 for that family (Corollary 4).

- *The cell coefficient in closed form* (Lemma 2), verified by exact polynomial algebra.

- *A certified nonzero cubic coefficient* (Proposition 2): a computer-assisted interval enclosure of $C_2'(0)$ for the test field.

- *Numerical confirmation* (Section 6.4): a deterministic test of the amplitude law, the residual ratios and the third-order stall, with the fixed-displacement family as a control.

- *Mixed-exponent operators* (Section 7): exact weights, first surviving moments and amplification factors for the sets $\{2, 3\}$, $\{2, 3, 4\}$, $\{2, 3, 4, 5\}$, with numerical confirmation that the $\{2, 3\}$ operator restores fourth order.

- *The spherical cap family* (Section 8): exact spherical triangle areas over a regular geodesic boundary polygon, which is not a flat-triangle rule, with an elementary parity proof of its even-power expansion.

- *An independent direct proof* for uniform latitude–longitude chordal cells (Proposition 3); the statement itself also follows from the classical theorem.
- Reproducible high-precision tables for the spherical test families, and the observation that the octant case reproduces a published 1994 tableau (Remark 1).

Classical background. Richardson–Romberg extrapolation [14, 3]; the Euler–Maclaurin expansion for quadrature over triangles and its extrapolation [11]; the flat-triangle approximation of curved surfaces, its extrapolation and its even-power expansion [8, 9, 17, 12]. By Corollaries 1–3 below, the even-power structure of the chordal spherical triangle, the fan-refined spherical polygon and the uniform latitude–longitude cell belongs to this classical material.

Not claimed. No new extrapolation method: the mixed-exponent operators are Richardson extrapolation for a general exponent sequence [4, 15]. No priority for extrapolating flat-triangle surface approximations. No claim that the degradation of high-order behaviour under perturbations proportional to the mesh size is new (see below). No even-power theorem for independent node jitter, adaptive meshes or arbitrary spherical meshes.

Related work on perturbed meshes. That extrapolation needs an asymptotic expansion, and that such expansions are tied to uniform or smoothly mapped meshes, is classical [13, 2]. Node perturbations proportional to the mesh size are a standard device for constructing irregular test grids in the verification of finite-volume schemes, where grids that cannot be smoothly mapped to a regular one are treated as a separate class [7]; the trapezoidal rule with nodes displaced by a fraction of the spacing has been analysed for convergence [1]. We did not find, in a limited search, the specific statements proved here for chordal area: the displacement-uniform expansion, the closed-form cell coefficient, and the exact cubic term with a certified nonzero coefficient. They are offered as new to the author's knowledge only; given the joint expansion, the cubic term itself is a one-line Taylor step.

3 N-Series scale operators

Put $h = N^{-1}$, let $TA(h) = A(h/2)$, and set $A_N = A(h)$. If $A(h) = A + c_2h^2 + c_4h^4 + c_6h^6 + O(h^8)$, define

$$\widehat{A}_N^{(2)} = \frac{4A_{2N} - A_N}{3}, \tag{1}$$

$$\widehat{A}_N^{(3)} = \frac{A_N}{45} - \frac{4A_{2N}}{9} + \frac{64A_{4N}}{45}, \tag{2}$$

$$\widehat{A}_N^{(4)} = -\frac{A_N}{2835} + \frac{4A_{2N}}{135} - \frac{64A_{4N}}{135} + \frac{4096A_{8N}}{2835}. \tag{3}$$

Each set of weights sums to one and eliminates h^2 and, respectively, the additional powers h^4 and h^6. Hence, subject to a sufficiently long asymptotic expansion and nonzero next coefficient, the errors are of orders 4, 6, and 8. These are Richardson–Romberg extrapolation formulas in a geometric N-Series setting.

In terms of the shift T (which acts on a pure power h^p as multiplication by 2^{-p}), Eqs. (1)–(3) are the nested Romberg stages

$$\widehat{A}^{(r)} = \prod_{k=1}^{r-1} \frac{4^kT - I}{4^k - 1}\, A, \qquad r = 2, 3, 4.$$

For example, $(16T-I)(4T-I)/45 = (64T^2-20T+I)/45$ reproduces Eq. (2). On an h^p term, stage k contributes the factor $(4^k 2^{-p}-1)/(4^k-1)$, which vanishes exactly when $p = 2k$. This product form gives the residual moments used in Section 6.

4 The regularity hypothesis

For a fixed spherical domain D and a specified geometrical area rule, call a refinement family *even-asymptotic through order* $2r$ when, as $h \downarrow 0$ along the admissible refinement sequence,

$$A(h) - A(D) = \sum_{k=1}^{r} c_{2k} h^{2k} + O(h^{2r+2}), \tag{4}$$

with coefficients independent of the refinement level. Equation (4), rather than informal smoothness alone, is the operative hypothesis of the $2, 4, 6, \ldots$ ladder.

For the flat-triangle area rule, Theorem 1 below identifies a class for which (4) holds to every order: a parametrization that is fixed (independent of h), smooth and regular on a closed triangle, sampled on the uniform subdivision of that triangle. Outside that class, and for other area rules, even-power structure must still be proved for the particular geometry and rule or supported by convergence tests. The map should preserve orientation and describe the same target domain at every scale.

5 Background: the classical expansion for chordal triangulations

This section records known results and verifies their hypotheses for the constructions used later. Nothing in it is claimed as new except the routine verifications. Let $\sigma = \{(u,v) : u \geq 0,\ v \geq 0,\ u+v \leq 1\}$. For $n \geq 1$ the *regular subdivision* of σ consists of the $n(n+1)/2$ triangles with vertices $\left(\frac{i}{n}, \frac{j}{n}\right), \left(\frac{i+1}{n}, \frac{j}{n}\right), \left(\frac{i}{n}, \frac{j+1}{n}\right)$, $0 \leq i+j \leq n-1$, together with the $n(n-1)/2$ triangles with vertices $\left(\frac{i+1}{n}, \frac{j}{n}\right), \left(\frac{i+1}{n}, \frac{j+1}{n}\right), \left(\frac{i}{n}, \frac{j+1}{n}\right)$, $0 \leq i+j \leq n-2$: in all, n^2 congruent subtriangles. For a map $\phi : \sigma \to \mathbb{R}^3$ let $S_n(\phi)$ denote the sum, over these n^2 subtriangles, of the Euclidean area of the flat triangle whose vertices are the images under ϕ of the subtriangle's vertices.

Theorem 1 (Verlinden–Cools [17]; Lyness [12])**.** *Let $\phi : \sigma \to \mathbb{R}^3$ have continuous partial derivatives of all orders up to the boundary, with $\phi_u \times \phi_v \neq 0$ on σ. Then there are constants $c_1, c_2, \ldots$ such that*

$$S_n(\phi) \ \sim \ \iint_\sigma \|\phi_u \times \phi_v\| \, du\, dv + \sum_{k \geq 1} \frac{c_k}{n^{2k}} \qquad (n \to \infty), \tag{5}$$

the leading term being the area of the surface $\phi(\sigma)$.

This is the case $g \equiv 1$ of Theorem 1 of [17], whose operator weights each flat-triangle area by a smooth function g and which also yields the corresponding expansions for Georg's modified trapezoidal and midpoint rules for $\int_S f\, dS$. It is Conjecture 5.1 of [8], restated as Conjecture 1 of [9]. Lyness [12] obtains the same h^2-expansion (his Theorem 5.6) with finite smoothness and a correspondingly truncated expansion. The expansion is asymptotic; no convergence of the infinite series is asserted. The statement requires ϕ to be the same map at every n.

Lemma 1 (Radial projection of a flat triangle)**.** *Let $A, B, C \in \mathbb{R}^3$ be linearly independent, put $P(u,v) = (1-u-v)A + uB + vC$ and $\phi = P/\|P\|$. Then $P \neq 0$ on σ, ϕ is real-analytic on a neighbourhood of σ, and*

$$\phi_u \times \phi_v = \frac{\det[A,B,C]}{\|P\|^4}\, P, \qquad \|\phi_u \times \phi_v\| = \frac{|\det[A,B,C]|}{\|P\|^3} \neq 0. \tag{6}$$

Proof. $P(u,v)$ is a combination of A, B, C with coefficients summing to one, so it cannot vanish when they are independent; hence ϕ is analytic near σ. Write $a = B - A$, $b = C - A$ and let $\Pi = I - \phi\phi^{\mathsf{T}}$. Then $\phi_u = \Pi a/\|P\|$ and $\phi_v = \Pi b/\|P\|$. For vectors projected onto the plane orthogonal to the unit vector ϕ, $\Pi a \times \Pi b = \big(\phi \cdot (a \times b)\big)\phi$, and $P \cdot (a \times b) = A \cdot \big((B - A) \times (C - A)\big) = \det[A, B, C]$ because the terms in u and v vanish. Substituting gives (6). □

When A, B, C are unit vectors, the radial projection carries the edges of the flat triangle to great-circle arcs, so $\phi(\sigma)$ is the spherical triangle ABC, and the map $(i, j) \mapsto \phi(i/N, j/N)$ is exactly the normalized-barycentric node set used in the experiments.

Corollary 1 (Chordal spherical triangle)**.** *Let A, B, C be linearly independent unit vectors and let A_N be the sum of the Euclidean areas of the N^2 chordal triangles of the normalized-barycentric triangulation of the spherical triangle ABC. Then, with $h = 1/N$, the family is even-asymptotic to every order:*

$$A_N \;\sim\; \mathrm{Area}(ABC) + \sum_{k \geq 1} c_k h^{2k}.$$

Proof. Lemma 1 shows that $\phi = P/\|P\|$ satisfies the hypotheses of Theorem 1, and $A_N = S_N(\phi)$. □

Corollary 2 (Fan-refined spherical polygon)**.** *Let $v_0, \ldots, v_{m-1}$ be unit vectors such that each triple (v_0, v_i, v_{i+1}), $1 \leq i \leq m-2$, is linearly independent, and let A_N be the sum over i of the chordal areas of Corollary 1 for the triangles $v_0 v_i v_{i+1}$. Then A_N has an asymptotic expansion in even powers of $h = 1/N$ whose constant term is the sum of the areas of the spherical triangles $v_0 v_i v_{i+1}$. That constant equals the area of the polygonal domain D exactly when those triangles have disjoint interiors and union D.*

Proof. A finite sum of expansions of the form (5). □

Corollary 3 (Uniform latitude–longitude cell)**.** *Fix $-\pi/2 < \phi_0 < \phi_1 < \pi/2$ and $\lambda_0 < \lambda_1$. Subdivide the parameter rectangle into a uniform $N \times N$ grid, split every cell along its (p_{00}, p_{11}) diagonal, and let A_N be the total Euclidean area of the resulting chordal triangles. Then A_N has an asymptotic expansion in even powers of $h = 1/N$ with constant term $(\lambda_1 - \lambda_0)(\sin\phi_1 - \sin\phi_0)$.*

Proof. Cut the rectangle along its own (p_{00}, p_{11}) diagonal into two triangles. The uniform grid with consistently oriented cell diagonals is the image of the regular subdivision of σ under the affine parametrization of each of the two triangles: the first family of subtriangles maps to the cell triangles (p_{00}, p_{10}, p_{11}) and the second to the cell triangles (p_{00}, p_{11}, p_{01}) for one large triangle, and conversely for the other. Composing with the spherical coordinate map, which is analytic with nonvanishing normal $\cos\phi \neq 0$ on the closed band, gives two maps satisfying Theorem 1. □

Remark 1 (The 1994 experiment of Georg and Tausch). Section 3 of [9] integrates $f = x^2 + y^2 + z^2$ over the positive octant of the unit sphere, parametrized by radial projection of the triangle with vertices ae_1, be_2, ce_3, on meshes $h_i = 2^{-i}$, and forms a Romberg tableau. Since $f = 1$ at every mapped vertex, their modified trapezoidal rule with $a = b = c = 1$ is the chordal area A_N of the octant, which is the $\beta = 90°$ case of Section 10. Table 1 confirms that the two computations agree to the digits printed in 1994. Their tableau also shows the h^2, h^4, h^6 columns. The octant experiment of this manuscript is therefore a higher-precision repetition of a published one.

N	$\lvert A_N - \pi/2\rvert$, this archive	Georg–Tausch (1994), trapezoidal, $a = b = c = 1$
4	7.75×10^{-2}	7.7×10^{-2}
8	2.03×10^{-2}	2.0×10^{-2}
16	5.14×10^{-3}	5.1×10^{-3}
32	1.29×10^{-3}	1.3×10^{-3}

Table 1: Raw chordal-area error for the octant. Left: recomputed on November 10, 2026 from the archived routine at 50 digits. Right: first column of the tableau printed in [9].

6 Main result: scale-dependent perturbations and the cubic obstruction

In the original perturbation experiment, boundary nodes of a latitude–longitude cell (Section 9) remained fixed, while interior nodes were displaced by a deterministic smooth field. Perturbation amplitude was specified as a fraction of local grid spacing; three seeds were used at each level. Because the amplitude is a fraction of the local spacing, the parametrization changes with N and Theorem 1 does not apply.

Relative perturbation	Raw	Two-scale	Three-scale	Four-scale
0.00	1.99996836	3.99999637	5.99994223	7.99911861
0.02	1.99997529	4.02004815	4.53899093	2.61899119
0.05	2.00001207	3.44420303	5.68235637	2.77050496
0.10	2.00014360	3.89815337	8.94059560	3.59334301
0.20	2.00066838	3.90830154	6.15480240	4.36890149

Table 2: Original perturbed-grid measurements (Table 6 of the June 2026 manuscript): aggregate final-level mean observed orders. These are diagnostics, not verified asymptotic exponents.

An apparent order such as 8.94059560 is not evidence of asymptotic convergence above the theoretical order: accidental coefficient cancellation, small denominators in measured error ratios, sign changes, or numerical floors can distort finite-level slopes. The original data establish only that the second-order raw behavior is comparatively robust and that higher-order extrapolation is not uniform in this test window.

Qualitative reading of Table 2. At small perturbation the two-scale order stays near 4 (0.02: 4.02) while the three- and four-scale orders fall well below their nominal 6 and 8, with four-scale values between about 2.6 and 4.4. This is qualitatively what a residual h^3 term would produce: the higher-order operators remove the h^4, h^6 terms that otherwise mask it, and the residual factors of Proposition 1 below (e.g. $-1/6$ versus $-1/810$) are applied to a coefficient that is linear in the amplitude. This is compatibility, not confirmation. The non-monotone entries, the erratic three-scale values, and the demonstrable pre-asymptotic variation in the unperturbed four-scale baseline (cf. Table 5, 30°) mean that the table cannot by itself separate a cubic residual from finite-window or round-off effects.

6.1 Conditional appearance of a cubic term

The key distinction is between a fixed nonuniform parametrization and a parametrization whose displacement amplitude changes with h. Suppose $A(h,\delta)$ is an approximation defined using a perturbation parameter δ, and the following smooth, uniform in δ even-power expansion holds near $(h,\delta)=(0,0)$:

$$A(h,\delta) - A = C_2(\delta)h^2 + C_4(\delta)h^4 + O(h^6), \tag{7}$$

where C_2 is three times continuously differentiable (or has a locally Lipschitz second derivative) and C_4 is at least once continuously differentiable near zero.

Remark 2 (Status of the hypothesis). Let the displacement be δV for a fixed smooth field V on the parameter domain that vanishes on its boundary. For each fixed small δ, the displaced grid is the image of the uniform grid under the fixed diffeomorphism $\psi_\delta = \mathrm{id} + \delta V$, so Theorem 1 already gives an even-power expansion with the same constant term A. Theorem 2 below proves more: for uniform square-cell grids with a fixed diagonal the coefficients are C^∞ in δ and the remainder is uniform, so (7) holds in full for that family. The local cubic terms that must cancel in [9] cancel only by summation over a uniform subdivision of a fixed map, which is consistent with a cubic term reappearing when the map itself varies with h.

Proposition 1 (Scale-dependent perturbations can create odd powers)**.** *Assume* (7) *with a remainder uniform for* $|\delta| \le \delta_0$ *and set* $\delta = \varepsilon h$*, with fixed* ε*. Then*

$$A(h,\varepsilon h) - A = C_2(0)h^2 + \varepsilon C_2'(0)h^3 + \Big[\tfrac{1}{2}\varepsilon^2 C_2''(0) + C_4(0)\Big]h^4 + O(h^5). \tag{8}$$

If $\varepsilon C_2'(0) \neq 0$*, then the two-, three- and four-scale even-power operators* (1)–(3) *each leave a cubic residual*

$$-\tfrac{1}{6}\,\varepsilon C_2'(0)\,h^3, \qquad -\tfrac{1}{90}\,\varepsilon C_2'(0)\,h^3, \qquad -\tfrac{1}{810}\,\varepsilon C_2'(0)\,h^3,$$

respectively, so the extrapolated error is $O(h^3)$ *and is not improved beyond third order by adding scales.*

Proof. Taylor-expand $C_2(\varepsilon h)$ and $C_4(\varepsilon h)$ in (7); the h^3 coefficient is $\varepsilon C_2'(0)$. For a scale-doubling weight set $\{w_j\}$ the surviving coefficient is $\varepsilon C_2'(0)\sum_j w_j 2^{-3j}$. Direct summation gives

$$\begin{aligned}
(2):&\ -\tfrac{1}{3} + \tfrac{4}{3}\cdot\tfrac{1}{8} = -\tfrac{1}{6},\\
(3):&\ \tfrac{1}{45} - \tfrac{4}{9}\cdot\tfrac{1}{8} + \tfrac{64}{45}\cdot\tfrac{1}{64} = -\tfrac{1}{90},\\
(4):&\ -\tfrac{1}{2835} + \tfrac{4}{135}\cdot\tfrac{1}{8} - \tfrac{64}{135}\cdot\tfrac{1}{64} + \tfrac{4096}{2835}\cdot\tfrac{1}{512} = -\tfrac{1}{810}.
\end{aligned}$$

Equivalently, by the product form in Section 3 the stage factors at $p=3$ are $-\frac{1}{6}$, $\frac{1}{15}$, $\frac{1}{9}$, whose cumulative products are $-\frac{1}{6}, -\frac{1}{90}, -\frac{1}{810}$. □

Remark 3. The notation $A(h,\varepsilon h)$ alone does not establish the existence of a cubic error term. It might vanish if $C_2'(0)=0$, if symmetry forces cancellations, or if a different mesh/area discretization fails the joint expansion assumption. Moreover, a fixed physical displacement δ and an h-scaled displacement define different refinement families; only comparisons made with identical geometric rules and numerical precision isolate this effect.

6.2 The joint expansion as a theorem

Theorem 2 (Displacement-uniform even-power expansion). *Let $\Omega = [0,1]^2$, let X be a C^∞ map from a neighbourhood of Ω into $\mathbb{R}^3$ with $X_s \times X_t \neq 0$ on Ω, and let V be a C^∞ planar vector field on a neighbourhood of Ω that vanishes on $\partial\Omega$. Let $\delta_0 > 0$ be small enough that $\delta_0 \sup_\Omega \|DV\| < \frac{1}{2}$, and put $\psi_\delta = \mathrm{id} + \delta V$ and $F_\delta = X \circ \psi_\delta$ for $|\delta| \le \delta_0$. For $N \ge 1$ and $h = 1/N$, sample F_δ on the uniform $(N+1)\times(N+1)$ parameter grid, split every cell along its (p_{00}, p_{11}) diagonal, and let $A(h,\delta)$ be the sum of the Euclidean areas of the resulting chordal triangles. Let $A = \iint_\Omega \|X_s \times X_t\|\, ds\, dt$. Then for every $r \ge 1$ there are C^∞ functions $C_2, \dots, C_{2r}$ on $[-\delta_0, \delta_0]$ and a constant M_r such that, for all sufficiently large N,*

$$\sup_{|\delta|\le\delta_0} \Big| A(h,\delta) - A - \sum_{k=1}^{r} C_{2k}(\delta) h^{2k} \Big| \le M_r h^{2r+2}. \tag{9}$$

Moreover, with q_0, q_2 the cell coefficients of Lemma 2 for $F = F_\delta$,

$$C_2(\delta) = \iint_\Omega q_2(x,\delta)\, dx - \frac{1}{24} \iint_\Omega \Delta_x q_0(x,\delta)\, dx. \tag{10}$$

Proof. Step 1: a uniform family of regular maps. For $|\delta| \le \delta_0$ the estimate $\|\psi_\delta(x) - \psi_\delta(y)\| \ge (1 - |\delta| \sup \|DV\|)\|x - y\|$ shows that ψ_δ is injective, and $\det D\psi_\delta > 0$. Since $V = 0$ on $\partial\Omega$, ψ_δ fixes $\partial\Omega$ pointwise, so by injectivity no interior point is mapped to $\partial\Omega$. For an interior point x_0 the path $\delta' \mapsto \psi_{\delta'}(x_0)$, δ' between 0 and δ, starts at x_0 and never meets $\partial\Omega$; hence ψ_δ maps the interior of Ω into itself. The image of the interior is open, because ψ_δ is a local diffeomorphism. It is also relatively closed in the interior: if $y_n = \psi_\delta(x_n)$ with x_n interior and $y_n \to y$ for an interior point y, then by compactness of Ω a subsequence of (x_n) converges to some $x \in \Omega$, and $\psi_\delta(x) = y$ by continuity; x cannot lie on $\partial\Omega$, since then $\psi_\delta(x) = x \in \partial\Omega$ whereas y is interior, so y is the image of an interior point. The interior of Ω is connected, so a nonempty subset that is both open and relatively closed is the whole interior. Thus ψ_δ is a bijection of Ω, and the change of variables formula gives $\iint_\Omega \|(F_\delta)_s \times (F_\delta)_t\| = \iint_\Omega \|X_s \times X_t\| \circ \psi_\delta \det D\psi_\delta = A$ for every δ. The family F_δ is jointly C^∞ in (x,δ) on a neighbourhood of $\Omega \times [-\delta_0, \delta_0]$, and $\|(F_\delta)_s \times (F_\delta)_t\|$ is bounded below by a positive constant there.

Step 2: the paired cell is even in h. For a cell centre x put $p_{ab}(x,h) = F_\delta\big(x + ((2a-1)h/2, (2b-1)h/2)\big)$, $a, b \in \{0,1\}$, and

$$G(x,h,\delta) = \tfrac{1}{2}\|(p_{10} - p_{00}) \times (p_{11} - p_{00})\| + \tfrac{1}{2}\|(p_{11} - p_{00}) \times (p_{01} - p_{00})\|.$$

Each difference of corners is h times a jointly smooth vector function, so each cross product is $h^2 w_i(x,h,\delta)$ with w_i jointly smooth and $w_i(x,0,\delta) = (F_\delta)_s \times (F_\delta)_t \neq 0$. By compactness there is $h_0 > 0$ such that $w_i \neq 0$ for $|h| \le h_0$, so $g = G/h^2 = \frac{1}{2}(\|w_1\| + \|w_2\|)$ is jointly C^∞ on $\Omega \times [-h_0, h_0] \times [-\delta_0, \delta_0]$, the smooth extensions of X and V being used where a shifted corner lies outside Ω. Replacing h by $-h$ interchanges $p_{00} \leftrightarrow p_{11}$ and $p_{10} \leftrightarrow p_{01}$, which exchanges the two unoriented triangles; hence $g(x,-h,\delta) = g(x,h,\delta)$. Taylor's theorem in h therefore gives

$$G(x,h,\delta) = h^2\Big(q_0(x,\delta) + q_2(x,\delta)h^2 + \cdots + q_{2r}(x,\delta)h^{2r}\Big) + O(h^{2r+4}),$$

with jointly C^∞ coefficients, $q_0 = \|(F_\delta)_s \times (F_\delta)_t\|$, and a remainder uniform in (x,δ).

Step 3: midpoint summation. $A(h,\delta)$ is the sum of G over the N^2 cell centres $x_{ij} = ((i + \frac{1}{2})h, (j + \frac{1}{2})h)$, so the remainders of Step 2 contribute $O(h^{2r+2})$ uniformly. For a smooth function q

on Ω, applying the one-dimensional midpoint Euler–Maclaurin formula in each variable (see [6], or Lemma 1 of [17] with offset $\frac{1}{2}$) gives

$$h^2 \sum_{i,j=0}^{N-1} q(x_{ij}) = \iint_\Omega q\,dx - \frac{h^2}{24}\iint_\Omega \Delta q\,dx + \sum_{\ell=2}^{r} h^{2\ell} B_{2\ell}[q] + O(h^{2r+2}),$$

where each $B_{2\ell}[q]$ is the integral over Ω of a fixed linear differential expression in q, and the remainder is bounded by a constant times a finite number of sup-norms of derivatives of q. Applied to $q = q_{2k}(\cdot,\delta)$, the coefficients are C^∞ in δ (differentiation under the integral) and the remainders are uniform in δ (compactness). Collecting powers of h gives (9), the constant term $\iint q_0 = A$ by Step 1, and (10). □

Lemma 2 (Cell coefficient)**.** *Let F be a C^∞ map into $\mathbb{R}^3$ near x with $N = F_s \times F_t \neq 0$ at x, and let $G(x,h)$ be the paired-cell chordal area of Step 2 above. Then $G = h^2\big(q_0 + q_2 h^2 + O(h^4)\big)$ with*

$$q_0 = \|N\|, \qquad q_2 = \frac{R\cdot N}{48\,\|N\|} + \frac{(F_{st}\cdot N)^2\,\|F_s + F_t\|^2}{8\,\|N\|^3}, \tag{11}$$

where $R = (F_s - F_t)\times(\partial_s+\partial_t)^3 F + (\partial_s-\partial_t)^3 F \times (F_s + F_t)$ and all derivatives are evaluated at x.

Proof. Write $D_1 = \partial_s + \partial_t$, $D_2 = \partial_s - \partial_t$, and put $c = p_{11} - p_{00}$, $e = p_{10} - p_{01}$, $m = \frac{1}{2}(p_{10}+p_{01}) - \frac{1}{2}(p_{00}+p_{11})$. Then $p_{10} - p_{00} = m + \frac{1}{2}e + \frac{1}{2}c$ and $p_{01} - p_{00} = m - \frac{1}{2}e + \frac{1}{2}c$, so the two cross products are $(\frac{1}{2}e + m)\times c$ and $(\frac{1}{2}e - m)\times c$, and

$$G = \tfrac{1}{2}\|n+\mu\| + \tfrac{1}{2}\|n-\mu\|, \qquad n = \tfrac{1}{2}\,e\times c, \quad \mu = m\times c.$$

Taylor expansion along the two diagonals gives $c = hD_1F + \frac{h^3}{24}D_1^3F + O(h^5)$, $e = hD_2F + \frac{h^3}{24}D_2^3F + O(h^5)$ and $m = \frac{h^2}{8}(D_2^2 - D_1^2)F + O(h^4) = -\frac{h^2}{2}F_{st} + O(h^4)$. Since $D_2F \times D_1F = 2N$,

$$n = h^2 N + \tfrac{h^4}{48}R + O(h^6), \qquad \mu = -\tfrac{h^3}{2}\,F_{st}\times(F_s+F_t) + O(h^5).$$

For vectors with $\|\mu\| < \|n\|$, $\frac{1}{2}(\|n+\mu\| + \|n-\mu\|) = \|n\| + \big(\|\mu\|^2 - (\mu\cdot\hat n)^2\big)/(2\|n\|) + O(\|\mu\|^4/\|n\|^3)$ with $\hat n = n/\|n\|$. The first term is $h^2\|N\| + \frac{h^4}{48}R\cdot\hat N + O(h^6)$, $\hat N = N/\|N\|$. For the second, decompose $F_{st} = u_T + \gamma\hat N$ with u_T in the span of F_s, F_t and $\gamma = F_{st}\cdot\hat N$. Then $u_T\times(F_s+F_t)$ is parallel to $\hat N$, while $\hat N\times(F_s+F_t)$ is orthogonal to $\hat N$ with norm $\|F_s+F_t\|$, so the part of μ orthogonal to $\hat N$ has squared norm $\frac{h^6}{4}\gamma^2\|F_s+F_t\|^2 + O(h^8)$. Dividing by $2\|n\| = 2h^2\|N\| + O(h^4)$ gives the second term of (11). □

The identity (11) was also verified as an exact polynomial identity, with the Taylor coefficients of F through fourth order treated as 42 independent symbols (`scripts/q2_lemma_symbolic_audit.py`): the two leading normals equal N, the h^3 terms of the two triangles cancel, the h^4 coefficient equals q_2, and the fourth-order coefficients do not enter.

Corollary 4 (Cubic obstruction for smoothly deformed square-cell grids)**.** *In the setting of Theorem 2, fix $\varepsilon\in\mathbb{R}$ and take $\delta = \varepsilon h$. Then (8) holds, and the two-, three- and four-scale even-power operators leave the cubic residuals $-\frac{1}{6}$, $-\frac{1}{90}$, $-\frac{1}{810}$ times $\varepsilon C_2'(0)h^3$. If $\varepsilon C_2'(0)\neq 0$, each extrapolated error has exact asymptotic order three; if $C_2'(0) = 0$, no stall at third order follows.*

Proof. Theorem 2 with $r = 2$ gives (7) with C^∞ coefficients and a remainder uniform for $|\delta| \le \delta_0$, and $|\varepsilon h| \le \delta_0$ for large N. Apply Proposition 1. □

Proposition 2 (Certified cubic coefficient for the test field; computer-assisted)**.** *Let X be the unit-sphere latitude–longitude map of the cell $30^\circ \le \phi \le 60^\circ$, $0^\circ \le \lambda \le 30^\circ$ in normalized coordinates $(s,t) \in [0,1]^2$, and let $V(s,t) = \sin(\pi s)\sin(\pi t)\big(\cos(2t + \frac{3}{10}), \sin(3s + \frac{7}{10})\big)$. Then*

$$C_2(0) \in -0.011894236621031846346030 51 \pm 8.5 \times 10^{-27},$$
$$C_2'(0) \in -6.80779112122198 \times 10^{-4} \pm 2.0 \times 10^{-19}.$$

In particular $C_2'(0) < 0$, and by Corollary 4 every $\varepsilon \neq 0$ gives a family whose even-power extrapolations have exact order three.

Proof. The four partial derivatives of V are bounded on Ω by π, $\sqrt{\pi^2+4}$, $\sqrt{\pi^2+9}$ and π, so the operator norm of DV is at most its Frobenius norm $\sqrt{4\pi^2+13} < 7.25$. Hence $\delta_0 = 0.068$ is admissible in Theorem 2, since $0.068 \times 7.25 = 0.493 < \frac{1}{2}$. This covers every run of Section 6.4: the fixed displacements satisfy $|\delta| \le 0.04$ and the scale-dependent ones $|\delta| = |\varepsilon|/N \le 0.2/8 = 0.025$. Differentiate (10) under the integral at $\delta = 0$, using $F_\delta = X + \delta(X_s V_1 + X_t V_2) + O(\delta^2)$ in (11) and $\partial_\delta q_0|_{\delta=0} = \operatorname{div}(\|X_s \times X_t\|\, V)$. The resulting integrand is an explicit combination of trigonometric functions whose only denominators are powers of $\|X_s \times X_t\| = (\pi/6)^2 \cos\phi$, with $\cos\phi \ge \frac{1}{2}$ on the cell; it is analytic on a complex neighbourhood of Ω. The symbolic differentiation was carried out in SymPy and the double integral enclosed by nested certified integration in Arb ball arithmetic [10] (`scripts/c2prime_enclosure.py`, 128-bit working precision). □

Remark 4 (What the certificate rests on). Proposition 2 is a computer-assisted statement: it relies on Theorem 2, Lemma 2, and the correctness of SymPy's differentiation and of Arb's enclosures. Three non-rigorous cross-checks agree with it: tensor Gauss–Legendre quadrature of the same integrand in binary64 arithmetic reproduces both midpoints to about 10^{-17} (`scripts/c2prime_crosscheck_numpy.py`); formula (10), evaluated from single cells without any grid sum, reproduces the grid-sum values of $C_2(\delta)$ at $\delta = 0, \pm 0.02, \pm 0.04$ to about 10^{-13} (`scripts/check_joint_expansion.py`); and direct finite differences of grid sums in δ converge to the same value. The certificate concerns this field on this cell only.

6.3 Falsifiable cubic-coefficient test

A sharper test of Proposition 1 is the amplitude law, not merely a measured convergence order. Choose a single smooth, boundary-vanishing perturbation vector field V and a fixed chordal triangulation convention. Here $h = N^{-1}$ in normalized cell coordinates (equivalently, $h = H/N$ with $H = 1$); $\delta = \varepsilon/N$ is the dimensionless displacement amplitude multiplying V. At signed amplitude ε, displace interior parameter-grid nodes by δV; retain the same physical domain and check triangle orientation. With $e(h,\varepsilon) = A(h,\varepsilon h) - A$, fit, over several successively refined grids and at sufficiently high precision,

$$e(h,\varepsilon) = b_2(\varepsilon)h^2 + b_3(\varepsilon)h^3 + b_4(\varepsilon)h^4 + \cdots . \tag{12}$$

Under the joint expansion (7), the precise small-amplitude prediction is $b_3(\varepsilon) = \varepsilon C_2'(0)$, including $b_3(0) = 0$; a nonzero slope requires $C_2'(0) \neq 0$. Test symmetric amplitudes $0, \pm\varepsilon_1, \pm\varepsilon_2$ with at least two distinct nonzero magnitudes. Check that $[b_3(\varepsilon) - b_3(-\varepsilon)]/(2\varepsilon)$ stabilizes to one value, while the even-in-ε part of b_3 tends to zero; compare the inferred slope across refinement windows and increased precision. By Remark 2, the fixed-δ runs of such a test should show clean even-power behaviour; that is a prediction of the classical theorem and a useful control.

By Proposition 1, the predicted cubic coefficients of the extrapolated errors are

$$-\tfrac{1}{6}\,\varepsilon C_2'(0) \ \text{(two-scale)}, \qquad -\tfrac{1}{90}\,\varepsilon C_2'(0) \ \text{(three-scale)}, \qquad -\tfrac{1}{810}\,\varepsilon C_2'(0) \ \text{(four-scale)}.$$

These coefficients are predictions of the proposition, not fitted measurements. A numerical test must distinguish failure of the joint expansion, vanishing derivatives, and pre-asymptotic effects; observed convergence orders alone cannot prove the hypothesis.

6.4 Numerical confirmation on a specified deterministic test

The test of the preceding subsection was carried out on November 10, 2026 with two separately written archived scripts: `scripts/cubic_amplitude_test_mpfr.py`, in 160-bit MPFR arithmetic (about 48 significant digits, identical on every platform), and `scripts/cubic_amplitude_test.py`, in the platform's extended precision. The cell is $30^\circ \le \phi \le 60^\circ$, $0^\circ \le \lambda \le 30^\circ$, with a uniform $N \times N$ parameter grid in normalized coordinates $(s,t) \in [0,1]^2$ and the (p_{00}, p_{11}) diagonal. Nodes are displaced to $(s,t) + \delta V(s,t)$ with the smooth, boundary-vanishing field

$$V(s,t) = \sin(\pi s)\sin(\pi t)\,\big(\cos(2t+0.3),\ \sin(3s+0.7)\big).$$

There are no random seeds. All numbers below are from the MPFR run. The extended-precision run reproduces every one of them to the digits shown, except the three- and four-scale orders of the fixed-displacement control, where extended precision reaches its rounding floor. Boundary nodes are fixed by construction. Orientation was checked for each of the 14,679,680 triangles in the 67 runs: the signed area of the displaced triangle in the parameter plane stayed positive, with minimum 0.878 times its undisplaced value, and every chordal normal pointed outward. No triangle failed either check.

Control: fixed displacement. For $\delta \in \{0, \pm 0.02, \pm 0.04\}$ held fixed as N grows, the observed orders at $N = 32 \to 64$ are 2.0000, between 4.0000 and 4.0044, between 5.997 and 6.000, and between 7.998 and 8.003 for the raw, two-, three- and four-scale values. This is the even-power behaviour that Theorems 1 and 2 predict for a fixed map. Extrapolating $N^2(A_N - A)$ gives $C_2(0) = -0.011894236621$, in agreement with Proposition 2 to all twelve digits. A two-step central difference of the grid-sum values at $\delta = \pm 0.02, \pm 0.04$ gives $C_2'(0) \approx -6.806 \times 10^{-4}$; the certified value is

$$C_2'(0) = -6.80779\ldots \times 10^{-4},$$

the discrepancy in the fourth digit being the truncation error of that coarse difference.

Scale-dependent displacement. With $\delta = \varepsilon/N$, the cubic coefficient b_3 of (12) was estimated from the two-scale residual as $-6N^3\big(\widehat{A}_N^{(2)} - A\big)$, with one extrapolation step in $1/N$ between $N = 64$ and $N = 128$. Table 3 compares it with the prediction $\varepsilon C_2'(0)$ from the certified coefficient. The two agree to about three significant figures, the coefficient is odd in ε, and it doubles when ε doubles. At $N = 64$, $\varepsilon = 0.1$, the part of the two-scale residual that is odd in ε is 4.3269×10^{-11}, against the predicted $-\frac{1}{6}\varepsilon C_2'(0)h^3 = 4.328 \times 10^{-11}$.

Residual factors and the stall at third order. Proposition 1 predicts that the three- and two-scale residuals are in the ratio $(-1/90)/(-1/6) = 1/15$ and the four- and three-scale residuals in the ratio $(-1/810)/(-1/90) = 1/9$. Table 4 shows both ratios approaching those values, and the extrapolated orders at or approaching three: adding scales no longer raises the order, as the proposition states.

ε	fitted b_3	predicted $\varepsilon C_2'(0)$
0.1	-6.809×10^{-5}	-6.808×10^{-5}
-0.1	6.809×10^{-5}	6.808×10^{-5}
0.2	-1.3611×10^{-4}	-1.3616×10^{-4}
-0.2	1.3613×10^{-4}	1.3616×10^{-4}

Table 3: Amplitude law for the cubic coefficient (new in v2.4). Fitted values: 160-bit MPFR grid sums. Predicted values: certified coefficient of Proposition 2.

		Residual ratios		Observed orders, $N \to 2N$			
ε	N	three/two	four/three	Raw	Two	Three	Four
0.1	64	0.069889	0.111099	2.0001	2.9656	3.0001	3.0000
0.1	128	0.068233	0.111108				
0.2	64	0.066837	0.111141	2.0001	2.9982	2.9997	3.0000
0.2	128	0.066767	0.111118				
Predicted		0.066667	0.111111	2	3	3	3

Table 4: Scale-dependent family $\delta = \varepsilon/N$: ratios of the extrapolated errors and observed orders (new in v2.4; 160-bit MPFR).

What this does and does not establish. By Corollary 4 and Proposition 2, the third-order stall for this cell and field is a theorem, subject to the review noted in the limitations; the test above is its numerical illustration and an end-to-end check of the constants. The fitted and certified cubic coefficients agree to three or four significant figures, the remaining difference being the finite-N error of the fit. The result concerns one field on one cell. It does not reconstruct the original experiment of Table 2, whose displacement field and seeds are not available, so the anomalies of that table are explained in kind but not reproduced.

6.5 Status of the original numerical comparison

The previously reported numerical comparison of fixed-map and scale-dependent perturbations has not been independently reproduced from the original implementation and is omitted from this release; Section 6.4 reports a new, fully specified test instead. Proposition 1 is an algebraic implication of its stated joint expansion hypothesis; Theorem 2 verifies that hypothesis for smooth fixed-field deformations of square-cell grids, not for the original perturbation code.

7 Mixed-exponent operators: restoring the ladder

Suppose the approximation has an expansion

$$A(h) = A + \sum_{p \in P} c_p h^p + \text{higher-order terms},$$

where $P = \{p_1, \dots, p_m\}$ is a prescribed set of distinct positive exponents. Using $m+1$ refinement levels $h, h/2, \dots, h/2^m$, seek

$$\widehat{A}(h) = \sum_{j=0}^{m} w_j A(h/2^j), \qquad \sum_{j=0}^{m} w_j = 1, \qquad \sum_{j=0}^{m} w_j 2^{-jp_k} = 0 \quad (k = 1, \dots, m). \tag{13}$$

The associated Vandermonde system has a unique solution for distinct exponents; this is Richardson extrapolation for a general exponent sequence [4, 15], and the formulas below are worked instances of that classical process on a doubling ladder. If $P = \{2, 3\}$, the three-scale weights are

$$\widehat{A}^{\{2,3\}}(h) = \frac{A(h)}{21} - \frac{4A(h/2)}{7} + \frac{32A(h/4)}{21}. \tag{14}$$

For $P = \{2, 3, 4\}$ the four-scale formula is

$$\widehat{A}^{\{2,3,4\}}(h) = -\frac{A(h)}{315} + \frac{4A(h/2)}{45} - \frac{32A(h/4)}{45} + \frac{512A(h/8)}{315}. \tag{15}$$

For $P = \{2, 3, 4, 5\}$ the five-scale formula is

$$\widehat{A}^{\{2,3,4,5\}}(h) = \frac{A(h)}{9765} - \frac{4A(h/2)}{651} + \frac{32A(h/4)}{279} - \frac{512A(h/8)}{651} + \frac{16384A(h/16)}{9765}. \tag{16}$$

For a term $c_p h^p$ the residual coefficient is multiplied by the exact moment $M_p = \sum_{j=0}^{m} w_j 2^{-jp}$. Direct rational evaluation gives

$$M_4^{\{2,3\}} = \frac{1}{56}, \qquad M_5^{\{2,3,4\}} = -\frac{1}{960}, \qquad M_6^{\{2,3,4,5\}} = \frac{1}{31744}. \tag{17}$$

Thus, when the indicated next coefficient is nonzero and a sufficiently long expansion holds, the three formulas have leading residuals $c_4 h^4/56$, $-c_5 h^5/960$, and $c_6 h^6/31744$, respectively, and leading orders 4, 5, and 6. The sums of absolute weights are approximately 2.143, 2.429, and 2.585, respectively: these give modest worst-case amplification factors for independent bounded input errors, although severe cancellation can still demand high working precision. They require more refinement levels than the even-only operators to reach a comparable order, and do not guarantee improved wall-clock efficiency. Numerical conditioning and computational cost should be measured rather than presumed.

Numerical check on the perturbed family. In the test of Section 6.4, the operator (14) applied to the scale-dependent family with $\varepsilon = 0.1$ gives observed orders 4.008 ($N = 32 \to 64$), 4.006 ($N = 64 \to 128$) and 4.003 ($N = 128 \to 256$), against third order for every even-power operator on the same data. For $\varepsilon = 0.2$ the corresponding values are 3.171, 3.758 and 3.903, still rising toward four within this window; the larger amplitude has a longer pre-asymptotic range.

8 Spherical cap area: a geodesic-polygon family

This family is not a flat-triangle rule and is not covered by Theorem 1. For a cap with polar angle α on the unit sphere, $A_{\text{cap}} = 2\pi(1 - \cos\alpha)$. Join the north pole to adjacent vertices of the regular geodesic boundary polygon and use the exact spherical area of each triangle [16]. With $c = \cos\alpha$, $s = \sin\alpha$, and $h = 2\pi/N$, the oriented spherical triangle area for sufficiently small $h > 0$ is

$$E_\alpha(h) = 2\arctan\left(\frac{s^2 \sin h}{1 + 2c + c^2 + s^2 \cos h}\right). \tag{18}$$

(This follows from $\tan(E/2) = t\sin h/(1 + t\cos h)$ with $t = \tan^2(\alpha/2) = (1 - c)/(1 + c)$, after multiplying numerator and denominator by $(1 + c)^2/(1 + c)$ and using $1 - c^2 = s^2$.) For $0 < \alpha < \pi$

the denominator is positive near zero and even in h, while the numerator is odd. Therefore E_α is odd analytic near zero:

$$E_\alpha(h) = a_1 h + a_3 h^3 + a_5 h^5 + \cdots, \qquad a_1 = 1 - \cos\alpha.$$

Since $A_N = N E_\alpha(2\pi/N)$, the polygon area has an even-power error expansion, and its constant term $2\pi a_1 = 2\pi(1 - \cos\alpha)$ equals the cap area exactly. This is the one-dimensional mechanism familiar from polygonal approximation of the circle; it is a proof for this cap family only.

α	Raw	Two-scale	Three-scale	Four-scale
15°	1.99989583	3.99999065	6.00093925	7.99993479
30°	1.99996740	4.00041698	6.00000053	7.95025527
60°	2.00016296	4.00014936	6.00003853	7.99880645
75°	2.00023455	4.00028432	6.00028420	8.00026989

Table 5: Original cap experiment: observed orders formed by comparing $N = 128$ with $N = 256$ (the row label is the cap polar angle α). The 30° four-scale value 7.95025527 is reproduced at 80-digit precision and represents a pre-asymptotic deviation, not a precision floor.

9 Latitude–longitude cell area: an independent direct proof

For the unit-sphere cell $\lambda_0 \le \lambda \le \lambda_1$ and $\phi_0 \le \phi \le \phi_1$, the exact area is

$$A_{\text{cell}} = (\lambda_1 - \lambda_0)(\sin\phi_1 - \sin\phi_0).$$

The approximation uses a uniform latitude–longitude parameter grid and two Euclidean chordal triangles per parameter cell. By Corollary 3 this family is even-asymptotic to every order. Proposition 3 below, written before the author knew of [17, 12], proves the same statement directly; it is retained as an independent elementary argument that uses only cell parity and the midpoint Euler–Maclaurin formula on a rectangle.

Cell	Raw	Two-scale	Three-scale	Four-scale
Equatorial 30° × 30°	1.99996836	3.99999637	5.99994223	7.99911861
Midlatitude 30° × 30°	1.99998486	3.99996146	5.99998582	8.00085300
Wide equatorial 60° × 30°	1.99992674	3.99992210	6.00000864	8.00036842
Northern 15° × 60°	1.99994200	3.99996772	5.99996443	8.00007636

Table 6: Original latitude–longitude experiment: observed orders at $N = 64$.

Proposition 3 (Even-power expansion for uniform latitude–longitude chordal cells)**.** *Fix $-\pi/2 < \phi_0 < \phi_1 < \pi/2$ and $\lambda_0 < \lambda_1$. Subdivide the corresponding unit-sphere parameter rectangle into a uniform $N \times N$ grid, with every cell triangulated along its (p_{00}, p_{11}) diagonal, and let A_N be the sum of the unsigned Euclidean areas of its two chordal triangles. With $h = 1/N$, for every positive integer r there are coefficients $c_2, \ldots, c_{2r}$, independent of N, such that*

$$A_N = (\lambda_1 - \lambda_0)(\sin\phi_1 - \sin\phi_0) + \sum_{k=1}^{r} c_{2k} h^{2k} + O(h^{2r+2}). \tag{19}$$

The assertion concerns this fixed rectangular domain and uniform, consistently triangulated refinement; it does not extend to arbitrary perturbed grids.

Direct proof. Put $\Delta\phi = \phi_1 - \phi_0$ and $\Delta\lambda = \lambda_1 - \lambda_0$. A parameter cell has centre (x, y) and corners $(x \pm \Delta\phi\, h/2, y \pm \Delta\lambda\, h/2)$. For the fixed p_{00}–p_{11} diagonal, let $g(x, y, h)$ be the sum of the unsigned Euclidean areas of the chordal triangles (p_{00}, p_{10}, p_{11}) and (p_{00}, p_{11}, p_{01}). Changing the sign of h permutes the opposite corners; taken together, the two unsigned triangle areas are invariant under this relabelling. Their sum therefore satisfies $g(x, y, -h) = g(x, y, h)$ for real h sufficiently close to zero. Each triangle has area one half of the norm of a cross product of edge vectors analytic in h. After factoring h^2, each squared norm has a positive leading coefficient proportional to $(\Delta\phi\, \Delta\lambda \cos x)^2$; hence its positive square root is real analytic for small signed h, uniformly on the closed latitude band. Thus g/h^2 extends to an even real-analytic function of h, uniformly with respect to (x, y) on the fixed rectangle. In particular, for any fixed r,

$$g(x, y, h) = \sum_{k=1}^{r+1} \rho_{2k}(x, y) h^{2k} + O(h^{2r+4}), \qquad \rho_2(x, y) = \Delta\phi\, \Delta\lambda \cos x,$$

with a uniform remainder. Rotational invariance makes the coefficient functions independent of longitude y, although this is not required for the midpoint argument. Summing the remainder over $N^2 = h^{-2}$ cells gives $O(h^{2r+2})$. For each coefficient, the cell-centre sum is a composite midpoint quadrature on a fixed rectangle, and the Euler–Maclaurin formula for smooth coefficient functions has an expansion in even powers of h. Collecting terms yields (19). The leading midpoint integral is

$$\int_{\phi_0}^{\phi_1} \int_{\lambda_0}^{\lambda_1} \cos\phi \, d\lambda \, d\phi = (\lambda_1 - \lambda_0)(\sin\phi_1 - \sin\phi_0).$$

□

Remark 5 (Scope of the parity argument). The midpoint nodes and the fixed diagonal are part of the definition of the refinement family. The assertion is an asymptotic expansion to each finite order, not a claim that an infinite series converges. In particular it does not cover the scale-dependent interior-node perturbation in Section 6. The pairing of the two triangles of a cell is what makes the local term even here; for a single triangle family the local term is not even, which is the difficulty identified in [9] and resolved in [17, 12].

10 Illustration of the classical theorem I: chordal spherical triangle

Take $A = (0, 0, 1)$, $B = (1, 0, 0)$, and $C = (\cos\beta, \sin\beta, 0)$. This is a birectangular spherical triangle: the sides AB and AC are meridians and BC lies on the equator, so the angles at B and C are right angles and the angle at the pole A is β. Its spherical excess is $\beta + \pi/2 + \pi/2 - \pi = \beta$, so $A_{\text{exact}} = \beta$. Map a normalized barycentric triangulation to the sphere and sum the Euclidean areas of its chordal triangles. Here $\det[A, B, C] = \sin\beta \neq 0$ for $0 < \beta < \pi$, so Corollary 1 applies and the orders in Table 7 illustrate a proved expansion. As a check of (6), $\iint_\sigma \sin\beta \, \|P\|^{-3} \, du\, dv = \beta$ was confirmed numerically to 50 digits for the four tabulated angles.

11 Illustration of the classical theorem II: irregular spherical polygons

The original study considered fixed irregular convex spherical polygons triangulated from a root vertex. Each fan triangle was refined using a normalized barycentric chordal patch. One six-vertex

β	Raw	Two-scale	Three-scale	Four-scale
15°	1.99942072	3.99915618	5.99886117	7.99716985
30°	1.99938698	3.99908575	5.99880136	7.99761444
60°	1.99923969	3.99881849	5.99858915	7.99835976
90°	1.99885313	3.99827889	5.99785193	7.99808456

Table 7: Original chordal triangle experiment: observed orders at $N = 64$.

example had angular vertices $(4°, -8°)$, $(18°, 5°)$, $(15°, 28°)$, $(2°, 42°)$, $(-14°, 30°)$, and $(-10°, 4°)$; the reported reference area by spherical excess was 0.32506277789824081542630164 9532. By Corollary 2, an irregular but fixed boundary cannot prevent the ladder, provided each fan triangle is non-degenerate; what the boundary geometry controls is whether the fan tiles the intended region, that is, whether the constant term is the target area.

N	Raw	Two-scale	Three-scale	Four-scale
16	1.99929248	3.99269841	5.99184034	8.01301620
32	1.99982228	3.99817005	5.99798552	8.00310155
64	1.99995552	3.99954224	5.99949790	8.00076594

Table 8: Original fixed six-vertex polygon experiment.

Vertices	Raw	Two-scale	Three-scale	Four-scale
5	1.99993053	3.99981653	5.99991832	7.99955672
6	1.99993586	3.99979158	5.99991390	8.00029966
8	2.00001032	4.00140962	5.99925770	7.99912295

Table 9: Original aggregate final-level orders over deterministic polygon families and seeds.

Boundary stress sweep. The original polygon tests separated fixed irregular boundaries from changes in the mesh rule. In a further boundary stress sweep, five-vertex cases at radii 15° and 30° with jitter 0.2 and 0.5, and eight-vertex cases at jitter 0.2, retained near-$2, 4, 6, 8$ behavior. Eight-vertex jitter-0.5 cases did not converge to the same reference area, with raw errors nearly constant and observed orders near zero. Corollary 2 supports the earlier reading of these cases: each fan triangle still carries an even-power expansion, so a nearly constant error indicates that the fan's constant term differs from the reference area (orientation, nonconvexity, overlap, or a different limiting region), not a failure of the expansion. This diagnosis has not been confirmed case by case. For all test families, the domain, orientation, target area, and approximation rule must be specified unambiguously.

12 Limitations and validation priorities

The even-power structure is now established for every fixed-map family in this manuscript: by the cap parity argument, and by Theorem 1 for the chordal triangle, fan polygon and latitude–longitude families. The tabulated observed orders remain numerical illustrations attributed to the

original manuscript. The original tests use convex or otherwise well-behaved spherical domains with available reference areas. Adaptive, scale-dependent and highly irregular meshes, and fans that do not tile the intended domain, are outside the established scope. The hypothesis of Proposition 1 is proved (Theorem 2) only for uniform square-cell grids with a fixed diagonal, deformed by a fixed smooth field that vanishes on the boundary. Independent node-by-node jitter, which is not the image of a smooth field, and adaptive or topology-changing triangulations are not covered, and the nonvanishing of $C_2'(0)$ is certified for one field on one cell only. The proofs of Theorem 2 and Lemma 2 and the certificate of Proposition 2 were produced with AI assistance; they have been checked by exact computer algebra and by reruns on the author's machine, but have not had an independent human proof reading, and no claim of historical novelty is established by the limited literature search reported in Section 2. The numerical test of Section 6.4 has not been repeated on other fields or geometries. The numerical anomalies of the original perturbed-grid Table 2 are *not* reproduced by this revision, because the original field and seeds are unavailable; the cubic mechanism explains them in kind only. Supplementary AI-assisted diagnostic fit tables requiring missing node and precision metadata are omitted from this release. Section 6.4 carries out items (i), (ii), (iv) and (v) of the following programme for a single field, and item (iii) in part; item (vi) is open. To strengthen the numerical result further, one should (i) test fixed-map and h-dependent-map meshes with the same node field, (ii) retain boundary constraints and triangle orientation checks, (iii) sweep several N and deformation amplitudes, (iv) fit the h^3 coefficient directly, (v) verify precision stability, and (vi) compare both error-versus-cost and observed-order-versus-cost across even-only and mixed-power operators.

13 Relation to the general N-Series framework

Spherical area supplies another setting in which geometrically defined approximants exhibit extrapolatable error modes. For flat-triangle approximations of smooth surfaces this was established three decades ago in the boundary-element literature; for those fixed-map families the present manuscript adds spherical specializations and examples, not mathematical novelty in the extrapolation or in the expansion. Its own contribution lies at the boundary of that theory: the cubic obstruction for scale-dependent families and the mixed-exponent operators that remove it. The episode also illustrates a general point for the N-Series programme: when a geometric approximant shows an even-power ladder under regular refinement, the Euler–Maclaurin literature for the relevant element shape should be checked before the ladder is described as conjectural. Williamson [18] surveys the regular spherical grids on which such area and quadrature questions arise in practice; Coxeter [5] is a general reference for the spherical geometry used.

14 Conclusion

The $2, 4, 6, 8$ ladder for spherical area is a property of fixed smooth refinement families, and for chordal triangulations it is classical. Its boundary is the subject of this manuscript. A displacement proportional to the grid spacing takes a family outside the classical theorem. For uniform square-cell grids deformed by a fixed smooth field we prove an even-power expansion that is uniform in the displacement, from which the scale-dependent family acquires a cubic term $\varepsilon C_2'(0)h^3$ that even-power extrapolation cannot remove; the two-, three- and four-scale operators leave it with exact factors $-1/6$, $-1/90$ and $-1/810$. For the test field the coefficient $C_2'(0)$ is given by an explicit integral and certified negative by interval arithmetic. A deterministic test confirms the amplitude law, the factor ratios $1/15$ and $1/9$, the stall of the extrapolated orders at three, and the recovery

of fourth order by the mixed-exponent operator for $\{2, 3\}$. Mixed-exponent operators with exact weights and moments are available whenever the exponents are identified. For fixed maps, the even-power expansion is proved here for spherical caps bounded by regular geodesic polygons, and follows from the theorem of Verlinden–Cools and Lyness, proving a conjecture of Georg, for chordal spherical triangles, fan-refined polygons and uniform latitude–longitude cells. The open items are an independent human reading of the new proofs and of the certificate, a fuller search of the literature on perturbed meshes, an extension from smooth fixed-field deformations to independent node jitter, tests on further fields and geometries, and a reconstruction of the original perturbed-grid experiment, whose anomalies are explained in kind but not reproduced.

A Reproducibility notes for the original experiments

The original June 2026 manuscript identifies the following standalone Python scripts:

- `spherical_area_nseries_cap_test.py`
- `spherical_triangle_nseries_area_test.py`
- `spherical_latlon_cell_nseries_area_test.py`
- `spherical_irregular_polygon_nseries_test.py`
- `spherical_irregular_polygon_batch_nseries_test.py`
- `spherical_irregular_polygon_stress_sweep_nseries_test.py`
- `spherical_perturbed_latlon_nseries_test.py`

For errors $e_N = |A_N - A_{\text{exact}}|$, the observed order is $p_N = \log(e_N/e_{2N})/\log 2$; the corresponding definition applies to the extrapolated errors. Some original experiment scripts were not included in the source/PDF review package, and their archived execution status is not established here. Numerical tables carried over from earlier versions are attributed to the original manuscript; this statement is not a reproducibility certification.

The test of Section 6.4 is `scripts/cubic_amplitude_test_mpfr.py` (requires `gmpy2`); it is deterministic, platform-independent, performs the orientation checks, and prints every number quoted there. Its output of November 10, 2026 is archived in `output/`. The earlier `scripts/cubic_amplitude_test.py` uses the platform's `long double` (x86 80-bit in the archived run) and serves as a second implementation. The statements of Section 6.2 are supported by `scripts/q2_lemma_symbolic_audit.py` (exact polynomial audit of Lemma 2; SymPy), `scripts/c2prime_enclosure.py` (certified enclosure of Proposition 2; SymPy and python-flint), `scripts/c2prime_crosscheck_numpy.py` (the same integrals by Gauss–Legendre quadrature; no interval library needed) and `scripts/check_joint_expansion.py` (formula (10) against grid sums). Their outputs of November 10, 2026 are archived in `output/`. The other checks added in this draft are in `scripts/check_v2_4_draft.py`: the Jacobian integral of Lemma 1 for the four tabulated triangles; the octant errors of Table 1, computed with the archived routine `chordal_triangle`; and exact rational evaluation of the weights (1)–(3) and (14)–(16), the cubic residual factors and the moments (17). The correspondence between the archived routine and the regular subdivision of Section 5 was established by reading the routine's index ranges.

B Revision history and provenance

Version 1 (June 3, 2026): five spherical area tests, tables of observed orders, cap parity argument, classical even-power scale operators, and limitations.

Version 2 (October 24, 2026): precise expansion-based regularity definition; conditional h^3 perturbation proposition; AI-assisted exploratory diagnostics, distinctly labeled; generalized mixed-power operators; stronger limits on what numerical observations establish. The original numerical values are preserved rather than silently replaced.

Version 2.1 (October 24, 2026): corrected the regularity needed for the $O(h^5)$ remainder; clarified provenance as exploratory AI-assisted review; added first surviving exact moments, error amplification factors, and a falsifiable signed-amplitude test of cubic-coefficient linearity.

Review supplement (October 24, 2026): added specified deformation fields, amplitude-fit sensitivity and provenance distinctions, plus an AI-assistance disclosure. The original perturbation experiment remains unverified.

Version 2.2 (October 25, 2026): stated the exact cubic residual factors $-1/6$, $-1/90$, $-1/810$ with a product-of-stages form of the operators; rewrote the cap formula numerator as $s^2 \sin h$ with a one-line derivation; described the test triangle as birectangular; described the cubic mechanism as consistent with, not established by, the perturbed-grid table; omitted supplementary fit data with incomplete provenance. No numerical table values were changed.

Version 2.3 release candidate (October 29, 2026): added Proposition 3 with its direct proof. Updated the abstract, scope and conclusion without changing the original numerical tables.

Version 2.4 draft (November 10, 2026): reorganized to lead with the manuscript's own results (the cubic obstruction, its numerical confirmation, the mixed-exponent operators and the cap family) and to present as background the literature on flat-triangle approximation of surface integrals [8, 9, 17, 12], which earlier versions did not cite. Added Theorem 1 (cited), Lemma 1, Corollaries 1–3, Remarks 1 and 2, Table 1, and the new numerical test of Section 6.4 with Tables 3 and 4. A later pass on the same day folded in a proof supplement: Theorem 2, Lemma 2, Corollary 4 and Proposition 2, with a related-work paragraph and the certified value of $C_2'(0)$ in place of a coarse finite-difference estimate (-6.806×10^{-4}) used in the first pass. No previously stated theorem, proposition, lemma or corollary was altered in either pass. The chordal triangle and fan-polygon families are reclassified from numerical evidence to instances of a classical theorem; Proposition 3 is reclassified from a new result to an independent proof of a known one. Earlier versions made no claim of a general theorem, so no previously stated result is withdrawn; what changes is attribution and status. Equation and table numbering no longer match earlier versions. No carried-over numerical table values were changed.

AI assistance in this revision. The relevant literature was identified, its hypotheses were compared with the archived implementation, the numerical test of Section 6.4 was designed, coded in two implementations and run, and this restructured draft was prepared with the assistance of Claude (Anthropic) at the author's direction. The proof of Theorem 2 was first drafted as a separate AI-assisted technical supplement at the author's direction and then audited and tightened in this session; the proof of Lemma 2, its symbolic audit and the interval enclosure were produced in this session. A separately written AI-assisted direct-grid computation corroborated the sign and magnitude of $C_2'(0)$. The author reran `c2prime_crosscheck_numpy.py` and `c2prime_enclosure.py` on his own Windows machine on November 10, 2026 and obtained the same values, including the certified sign. Both implementations of the numerical test came from the same AI-assisted session, so their agreement shows precision stability, not independence. The scripts are archived and deterministic so that the author and readers can rerun them; an independent rerun is pending. The

1994 papers [9, 17, 12] were read in full for this purpose; [8] was not available and is cited on the authority of the three papers that state and prove its Conjecture 5.1. This is an AI-assisted revision, not an independent human peer review, and the author's own review of this draft is pending.

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